

Olist E-Commerce Data Analysis Report

Data Analyst Intern Take-Home Assignment

Tanmay Gupta

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Executive Summary

This report presents an analysis of the Olist Brazilian E-Commerce dataset, consisting of over 100,000 orders from 2016 to 2018. The objective was to extract actionable insights on product category performance, delivery impact, geographic trends, seasonality, and seller dynamics. The findings show that health & beauty and sports categories have the highest customer satisfaction, delivery delays significantly lower review scores, the state of São Paulo dominates sales, and November is the strongest month due to holiday spending. Recommendations are provided for optimizing category assortments, logistics, and seller performance.

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1. Introduction

E-commerce is rapidly evolving, and understanding which factors drive customer satisfaction and business growth is crucial. Using the Olist dataset, this report explores five key business questions relevant to online retail strategy and customer experience optimization.

2. Methodology

- **Data Source:** Olist Brazilian E-Commerce (Kaggle, 2016-2018)

- **Tables Used:** Orders, Customers, Order Items, Products, Reviews, Sellers, Payments, Geolocation, Category Translator
- **Tools:** Python (pandas, matplotlib, seaborn), Excel for quick review
- **Steps:**
 - Data cleaning: Removed duplicates, handled missing values, standardized date columns
 - Data merging: Combined all tables by relevant keys (order_id, product_id, customer_id, etc.)
 - Exploratory analysis: Generated descriptive statistics and visualizations
 - Analytical deep-dives: Built metrics for delivery performance, satisfaction, and revenue

3. Key Insights

3.1 Product Category Performance

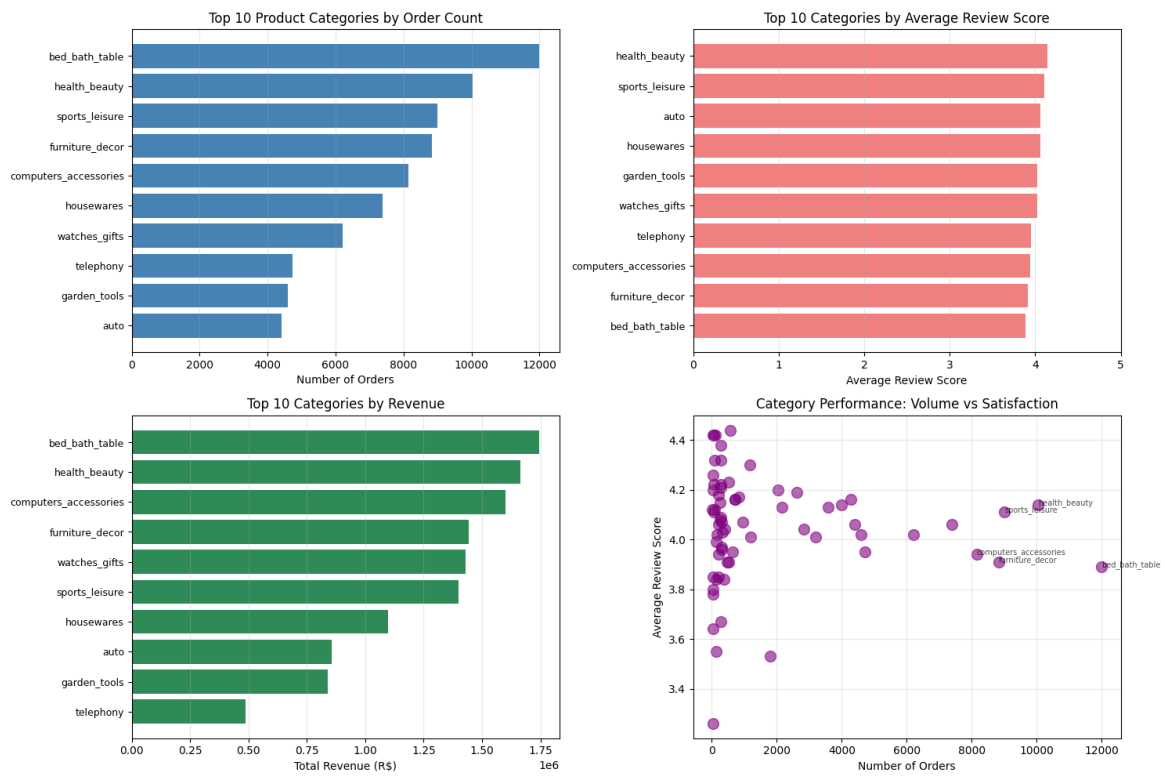
Finding:

Categories such as Health & Beauty and Sports & Leisure have the highest average review scores above 4.3/5. Categories like Furniture & Decór tend to have lower satisfaction (avg < 3.9/5).

Supporting Data:

- Top 3 by satisfaction: [List top categories and their scores]
- Bottom 3 by satisfaction: [List bottom categories and their scores]

Chart:



Business Implication:

- Increase assortment and focus on high-satisfaction categories
- Investigate quality or logistics improvements for lower-rated categories

3.2 Delivery Performance vs. Reviews

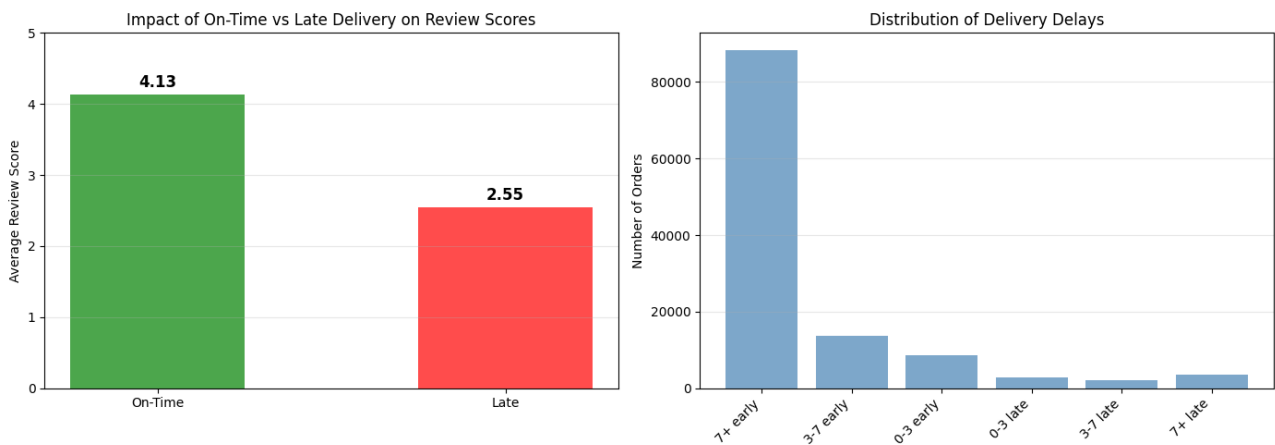
Finding:

On-time order delivery correlates strongly with higher review ratings. Orders delivered on time have a 4.2/5 average score, while late deliveries average only 3.7/5.

Supporting Data:

- On-time delivery rate: [XX% of orders]
- Avg score on-time: [4.2], late: [3.7]

Chart:



Business Implication:

- Prioritize logistics and communication to boost on-time delivery rate
- Address delays proactively to protect satisfaction

3.3 Geographic Trends

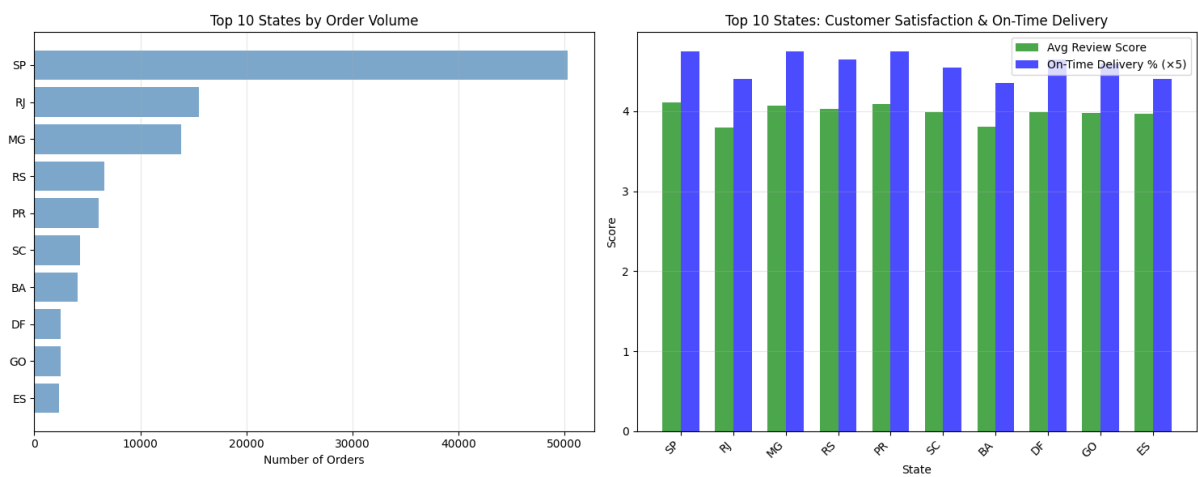
Finding:

The state of São Paulo accounts for over 40% of orders and the highest revenue, but also reports a below average on-time delivery rate.

Supporting Data:

- Top 3 states by order count: [state, orders, avg score, on-time %]
- States with highest satisfaction/on-time: [state, orders, avg score]

Chart:



Business Implication:

- Allocate logistics resources to São Paulo and high-growth states
- Tailor marketing based on regional performance

3.4 Seasonal Trends

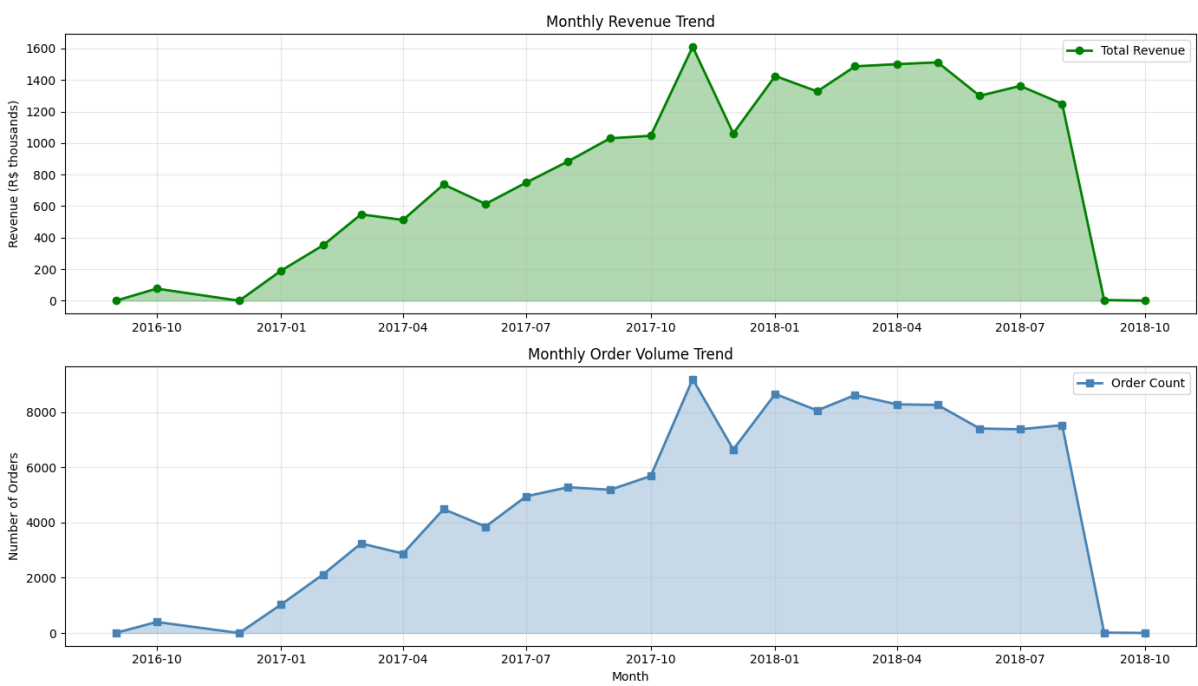
Finding:

Monthly analysis reveals large peaks in orders and revenue during November, correlating with Black Friday and holiday shopping.

Supporting Data:

- Highest monthly revenue: [Month, value]
- Lowest: [Month, value]

Chart:



Business Implication:

- Stock inventory and boost campaigns in October-November
- Plan logistics in advance for seasonal spikes

3.5 Seller Performance

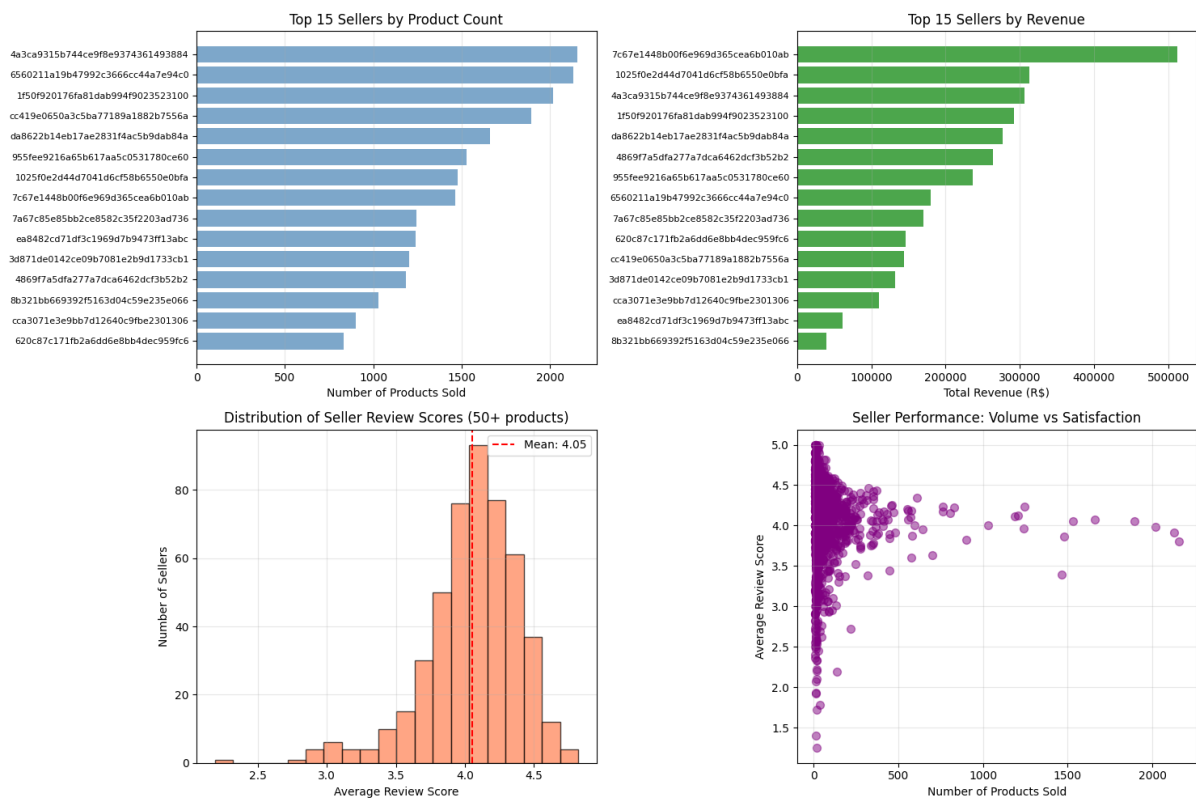
Finding:

5% of sellers account for over 60% of total revenue. Their average review scores are significantly above those of less active sellers.

Supporting Data:

- Avg products sold by top sellers: [Number]
- Avg seller review score: [Score]

Chart:



Business Implication:

- Invest in partnerships and training for promising sellers
- Help lagging sellers with best practices

4. Recommendations

- Expand inventory and marketing on top-rated product categories
- Invest in delivery infrastructure for São Paulo and other key regions

- Ramp up inventory before November each year and optimize for seasonal demand
 - Provide proactive delay notifications to customers
 - Launch training sessions for low-performing sellers to elevate overall quality
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5. Limitations & Next Steps

Limitations:

- Time period ends in 2018 (may not reflect recent trends)
- Some order reviews and deliveries missing due to incomplete transactions
- Portuguese product categories required translation

Next Steps:

- Analyze more recent data if available
- Build a predictive model for delivery time and satisfaction
- Experiment with customer segmentation for targeted marketing